ESG & Stock Returns Dataset

Data dictionary for the EAIF group assignment, Autumn Semester 2026

Overview

Source	Simulated by the course for this assignment.
Licence	Creative Commons Attribution 4.0 International (CC BY 4.0), EAIF course. Cite the dataset and its licence in the memo and on the slides.
Location	`data/esg2026/` in the course GitHub repository (`umatter/EDFB`); the notebook loads these files with `pandas.read_csv` from `https://github.com/umatter/EDFB/tree/main/data/esg2026`.
Files	`firm_year_panel.csv`, `esg_scores_restated.csv`, `stock_returns_monthly.csv`, `factors_monthly.csv`, `universe_current.csv` – five data files, described column by column below.
Grain and join keys	The panel, the restated scores and the universe file are keyed on `Company_ID` (the panel and the restated file also on `Year`); the returns and the factors file are keyed on `Month`. `Company_ID` is the join key everywhere a company appears.
Fiscal year and publication	Each row of the panel is one firm and one fiscal year t. Its `Publication_Date` is the date that fiscal year's row became public; nothing in that row is known before that date. The row for fiscal year t is public from `Publication_Date`, always between 31 March and 31 May of $t+1$.

Simulated data with real problems

There are no real firms behind these rows. The generator planted a small number of data problems of the kind real financial data has: things that are not what they seem, values that are known too early or too late, samples that are not what you think they are. Finding and treating them is task (a) and is graded. The generator also planted one genuine relationship between ESG scores and returns; whether you find it, and only it, is task (c).
Your memo must still contain a section on what a real-data version of your analysis would need beyond what you did here. That section is graded (see the assignment handout's rubric).

Files and columns

`firm_year_panel.csv`: one row per firm and fiscal year; fundamentals, point-in-time ESG scores, environmental metrics

14,250 rows, 18 columns.

Variable	Type	Unit	Range or values	Missing
Company_ID	integer		1 to 1,376, median 601	0
Company_Name	string		Firm_1 to Firm_999	0
Industry	string		Finance, Manufacturing, Energy, Retail, Consumer Goods, Technology, Utilities, Healthcare, Transportation	0
Region	string		Asia, Oceania, North America, Europe, Middle East, Latin America, Africa	0
Year	integer	fiscal year	2012 to 2025, median 2018	0
Publication_Date	string	YYYY-MM-DD	2013-03-31 to 2026-05-30	0
Revenue	float	USD million	30.3 to 63,993,116, median 2,406	0
Profit_Margin	float	percent	-5.47 to 29.34, median 10.58	0
Market_Cap	float	USD million	22.6 to 157,136,175, median 4,882	0
Growth_Rate	float	percent per year	-30.97 to 61.11, median 4.82	1,376
ESG_Score	float	0 to 100	4.6 to 96.8, median 55.1	0
Environmental_Score	float	0 to 100	0 to 100, median 55.25	0
Social_Score	float	0 to 100	0 to 100, median 54.5	0
Governance_Score	float	0 to 100	0 to 100, median 54.3	0
Carbon_Emissions	float	tons CO2	238.1 to 395,293,803, median 188,237	1,887
Water_Usage	float	cubic metres	189.5 to 465,141,673, median 104,937	1,887
Energy_Consumption	float	MWh	1,802 to 2,295,873,427, median 1,594,587	1,887
Analyst_Outlook	integer	1 (low) to 5 (high)	1 to 5, median 3	0

esg_scores_restated.csv: ESG scores as a current download would show them, one row per firm and fiscal year

14,250 rows, 6 columns.

Variable	Type	Unit	Range or values	Missing
Company_ID	integer		1 to 1,376, median 601	0
Year	integer	fiscal year	2012 to 2025, median 2018	0
ESG_Score	float	0 to 100	10.9 to 96.8, median 55.5	0
Environmental_Score	float	0 to 100	0 to 100, median 55.6	0
Social_Score	float	0 to 100	0 to 100, median 55	0
Governance_Score	float	0 to 100	0 to 100, median 55	0

stock_returns_monthly.csv: total stock return per firm and month

171,764 rows, 3 columns.

Variable	Type	Unit	Range or values	Missing
Company_ID	integer		1 to 1,376, median 594	0
Month	string	YYYY-MM	2012-01 to 2025-12	0
Return_Pct	float	percent per month	-95 to 118.52, median 0.87	0

factors_monthly.csv: risk-free rate, market excess return, size factor and one factor per industry

168 rows, 13 columns.

Variable	Type	Unit	Range or values	Missing
Month	string	YYYY-MM	2012-01 to 2025-12	0
Rf_Pct	float	percent per month	0.1 to 0.4, median 0.1	0
Mkt_Excess_Pct	float	percent per month	-13 to 10.94, median 1.02	0
SMB_Pct	float	percent per month	-4.84 to 6.54, median 0.18	0
Ind_Consumer_Goods_Pct	float	percent per month	-5.55 to 4.45, median -0.15	0
Ind_Energy_Pct	float	percent per month	-4.32 to 5.31, median -0.02	0
Ind_Finance_Pct	float	percent per month	-6.04 to 4.95, median -0.12	0
Ind_Healthcare_Pct	float	percent per month	-5.19 to 5.14, median -0.02	0
Ind_Manufacturing_Pct	float	percent per month	-6.35 to 6.9, median 0.46	0
Ind_Retail_Pct	float	percent per month	-5.61 to 5.32, median -0.29	0
Ind_Technology_Pct	float	percent per month	-5.42 to 4.71, median 0.39	0
Ind_Transportation_Pct	float	percent per month	-6.06 to 7.77, median -0.17	0
Ind_Utilities_Pct	float	percent per month	-5.85 to 4.4, median 0.09	0

`universe_current.csv`: firms in the universe at the end of the last year

987 rows, 4 columns.

Variable	Type	Unit	Range or values	Missing
Company_ID	integer		1 to 1,376, median 716	0
Company_Name	string		Firm_1 to Firm_999	0
Industry	string		Finance, Manufacturing, Energy, Retail, Consumer Goods, Technology, Utilities, Transportation, Healthcare	0
Region	string		Asia, Oceania, Europe, Middle East, North America, Africa, Latin America	0

Counts

Fiscal year	Firm-year rows
2012	1,000
2013	1,028
2014	1,038
2015	1,039
2016	1,030
2017	1,023
2018	1,019
2019	1,010
2020	1,016
2021	1,012
2022	1,026
2023	1,019
2024	1,003
2025	987

Industry	Firms
Consumer Goods	161
Energy	127
Finance	177
Healthcare	166
Manufacturing	164
Retail	135
Technology	137
Transportation	152
Utilities	157

Region	Firms
Africa	114
Asia	290
Europe	294
Latin America	138
Middle East	118
North America	284
Oceania	138

Conventions

- **Units.** USD million (`Revenue`, `Market_Cap`); percent (`Profit_Margin`); percent per month (`Return_Pct` and the factor columns); tons CO_2 (`Carbon_Emissions`); cubic metres (`Water_Usage`); MWh (`Energy_Consumption`); scores 0 to 100 (`ESG_Score` and its three components).
- **Returns** in `stock_returns_monthly.csv` are total returns, in percent per month.
- **The risk-free rate** is in `factors_monthly.csv` (`Rf_Pct`), together with the market excess return, a size factor, and one factor per industry (named by industry, for example `Ind_Technology_Pct`).
- `Company_ID` is the join key everywhere a company appears, across all five files.
- **The panel is unbalanced:** firms enter and leave. Not every `Company_ID` has a row in every fiscal year, and not every firm that appears in the panel is in `universe_current.csv`.
- `Growth_Rate` is missing in a firm's first fiscal year in the panel (there is no prior year to compute growth from).

What this file does not tell you

The files contain data problems of the kind real data has. This dictionary documents every column as it appears; it does not point them out. Finding them is task (a).

Attribution

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